NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Exploratory Data Analysis

Amazing Creative Desk Coworking – customer segmentation

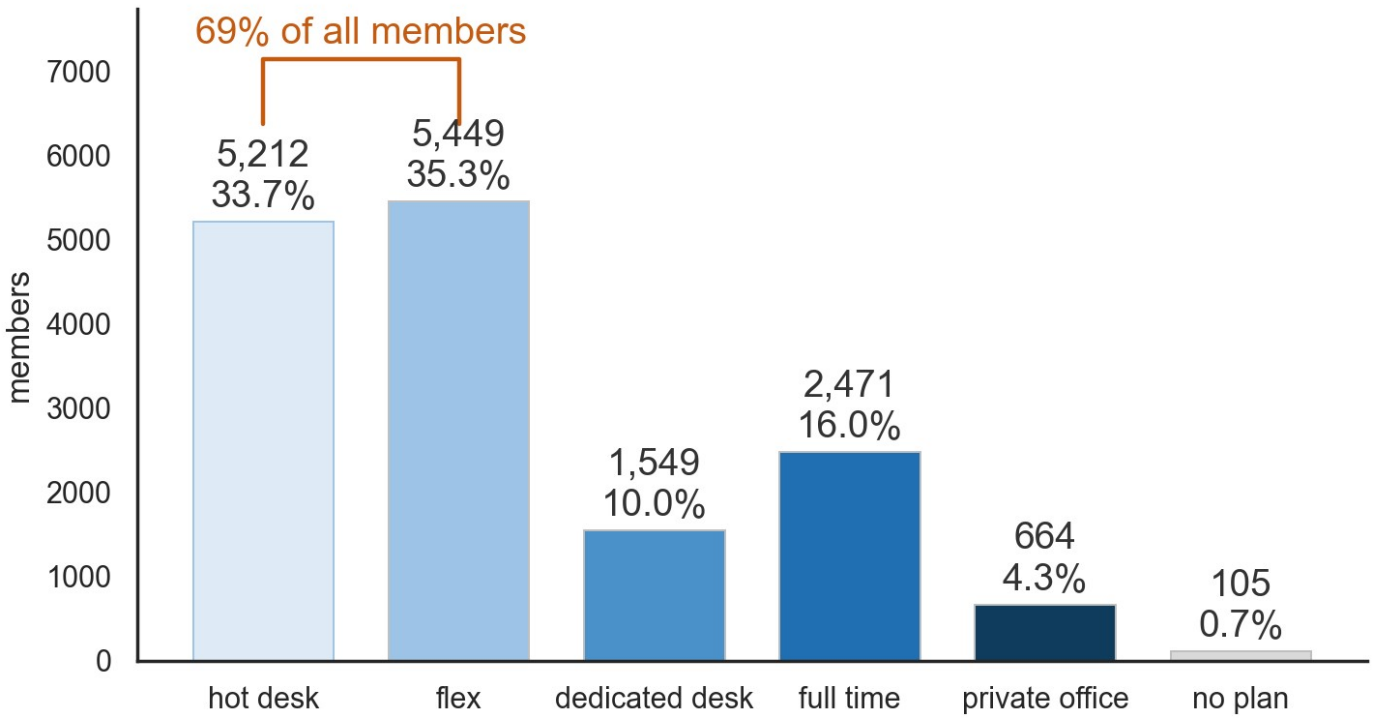
Group 10

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Two thirds of the members sit on the two cheapest plans



Members crowd into the two cheapest plans, and a small group has no plan recorded.

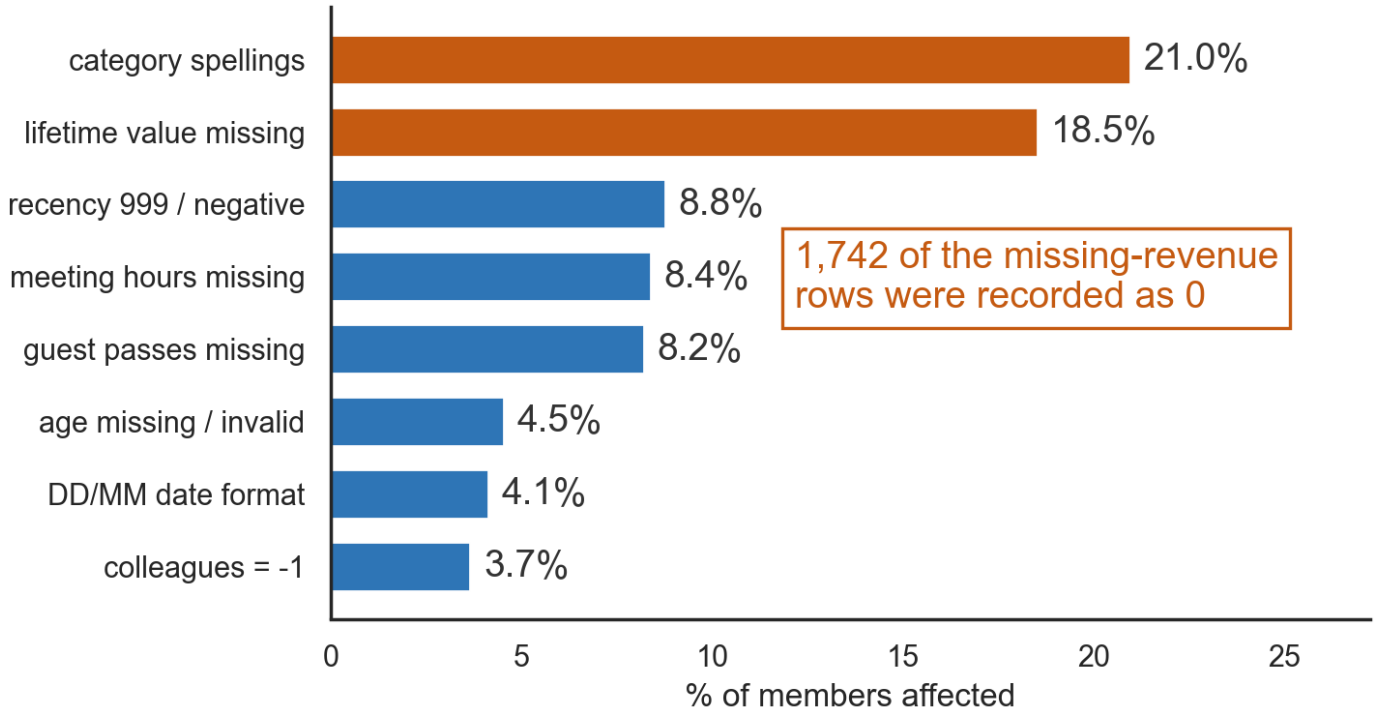
ACDC is a coworking operator; we are preparing a customer segmentation. Three years, 15,465 rows, 15 duplicates dropped, 15,450 members, 31 variables.

- 69% of members are on those two plans, so a plan-based segmentation mostly describes them.
- 105 members have no plan, so we cannot lean on it alone.

Source: EDA notebook

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We fixed spelling and flagged gaps instead of filling them in



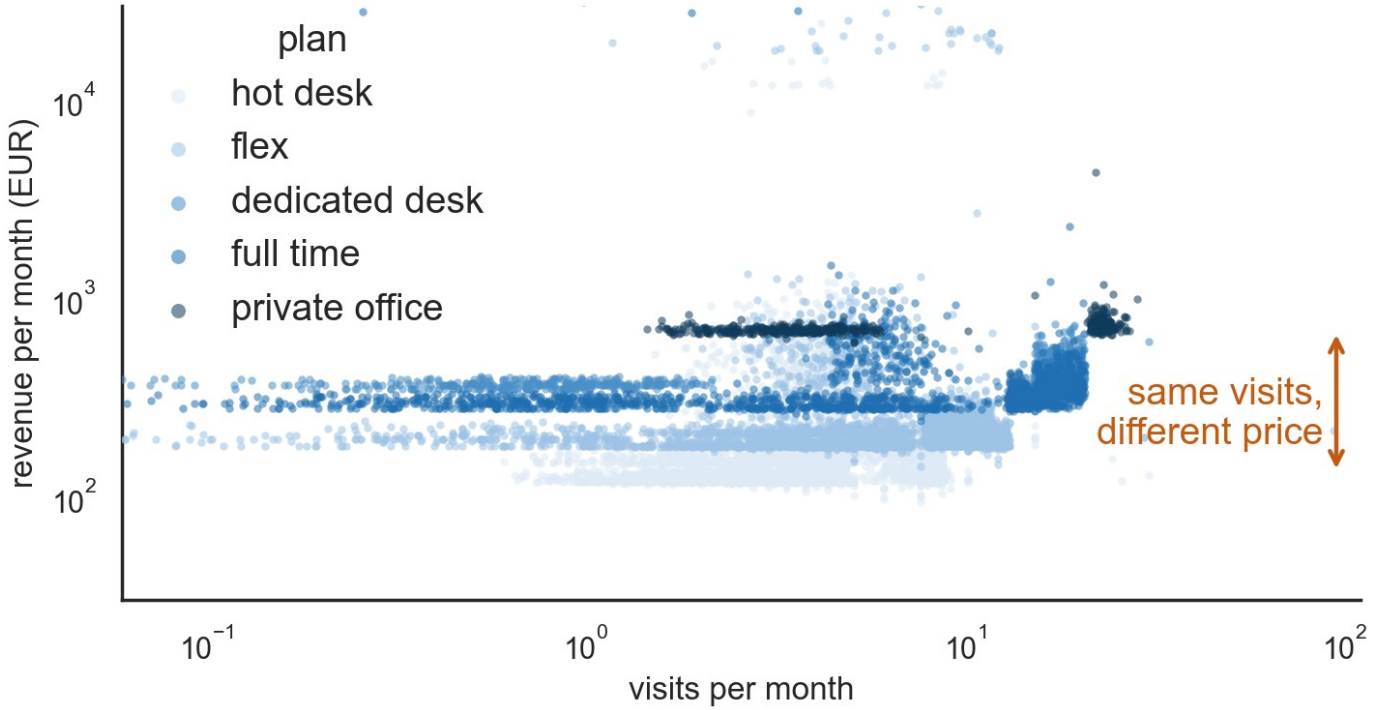
- 3,238 rows spelled the same category in different ways, so we mapped them back.
- 1,742 members showed zero revenue, 263 of them on a dedicated desk or private office, so we read zero as unknown.
- 408 rows had counts that disagreed with their plan, so we flagged them instead of guessing.

Two issues reach about a fifth of the base each, so they decide how far the rest can be trusted.

Source: EDA notebook

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Revenue per month follows the plan, so it adds little to a segmentation



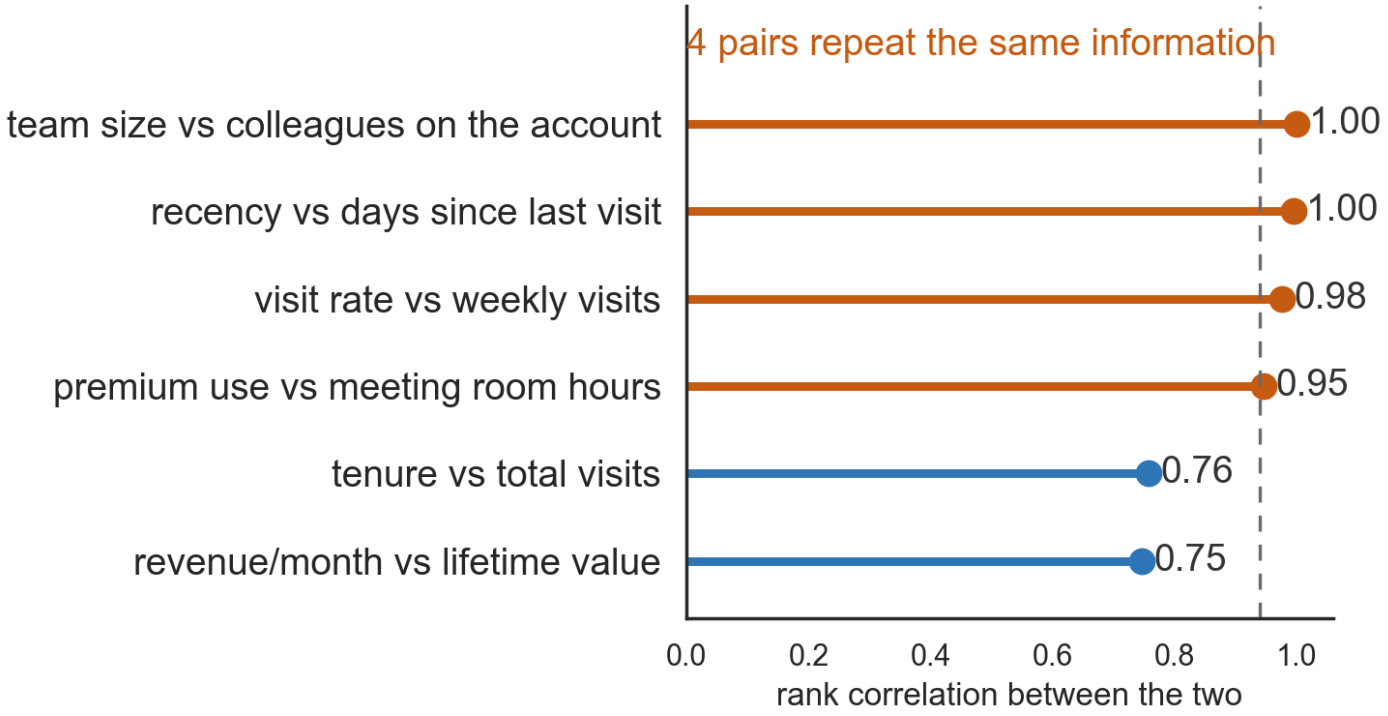
- Each plan sits at its own revenue level, so revenue mostly restates the price list.
- Visits do not follow the plan: every plan spans the same range of them.
- So we look for segments in how members use the space, not in what they pay.

Members on the same plan pay nearly the same amount but turn up at very different rates.

Source: EDA notebook

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We shrank the tails and dropped repeated features before clustering



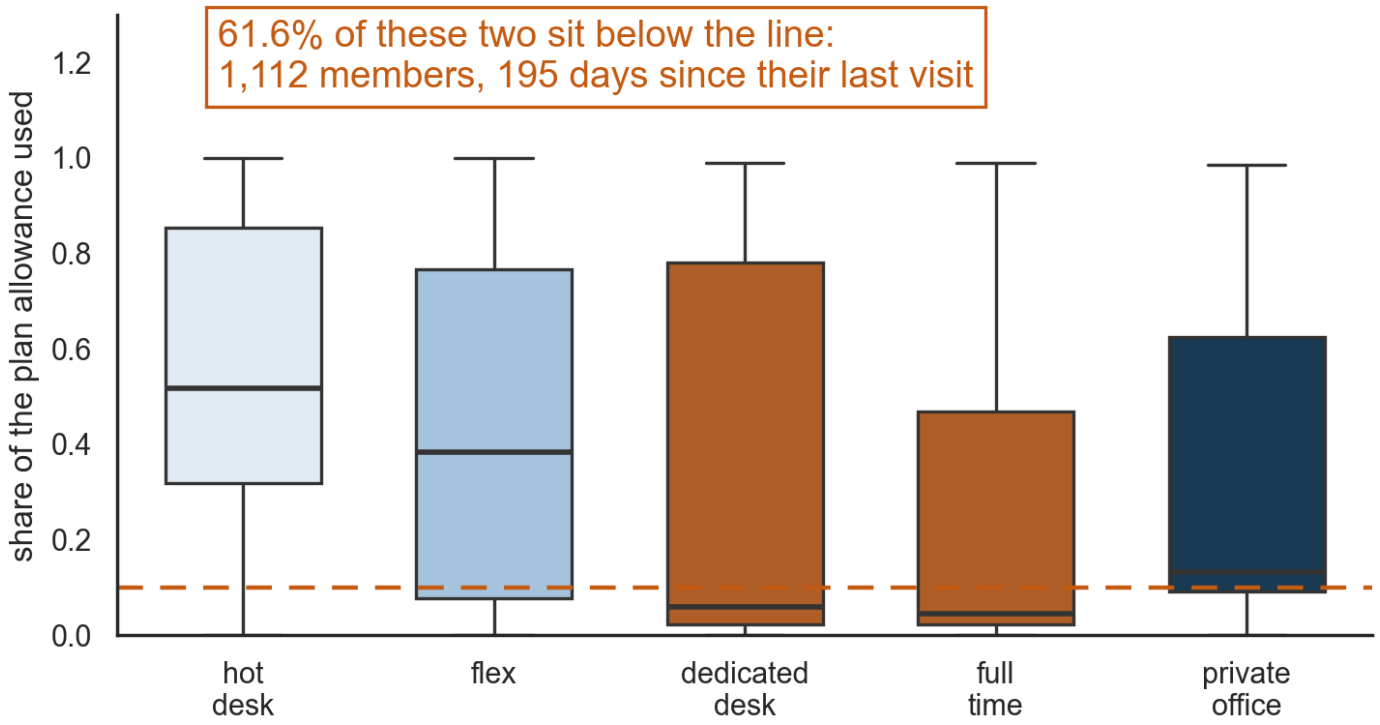
- We took logs first, because the top 1% of members hold 45.8% of all recorded revenue.
- Four pairs of features said the same thing, so we keep one of each.
- Unusual members stay in, because they carry real information, and gaps stay flagged.

Four pairs sit above the line, so keeping both would count the same evidence twice.

Source: EDA notebook

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Most active desk members barely use the desk they pay for



Members who left had used the space, so the low-use group is the one still paying for it.

- 1,112 active members on a dedicated desk or full-time plan use under a tenth of their allowance.
- 74.4% of the base has already ended, so clusters may only split active from ended members.
- We will compare those two groups separately and expect 4–6 clusters.

Source: EDA notebook